

计算系统生物学考试

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2026 年 6 月 13 日

1. Chemical kinetics of cAMP signaling

a. Based on the equations given, the equations giving the time evolution of $[AC]$ and $[IC]$ are as follows

$$\frac{d[AC]}{dt} = k_A^+[A][C] - k_A^-[AC] \quad (1)$$

$$\frac{d[IC]}{dt} = k_I^+[I][C] - k_I^-[IC] \quad (2)$$

Since $[A] = [A_{tot}] - [AC]$, if we assume $[AC] \ll [A_{tot}]$ (hence $[A] \approx [A_{tot}]$), we can rewrite the first equation as

$$\frac{d[AC]}{dt} \approx k_A^+[A_{tot}][C] - k_A^-[AC] = k_A^- \left(\frac{k_A^+}{k_A^-} [A_{tot}][C] - [AC] \right) \quad (3)$$

Denote $[C]$ as c , we define $\tilde{a} = \frac{k_A^-}{k_A^+[A_{tot}]}[AC]$. Then we can rewrite the equation above as

$$\frac{d\tilde{a}}{dt} = k_A^- (c - \tilde{a}) \quad (4)$$

Since a and $\tilde{a}$ are only different in terms of the unit of concentration, we have

$$\frac{da}{dt} = k_A^- (c - a) \quad (5)$$

Similarly, we can define $\tilde{i} = \frac{k_I^-}{k_I^+[I_{tot}]}[IC]$ and rewrite the second equation as

$$\frac{d\tilde{i}}{dt} = k_I^- (c - \tilde{i}) \quad (6)$$

b. At the rapid equilibrium, we have

$$k_R^+ a^2 [R] = k_R^- i^2 [R^*] \quad (7)$$

By $[R] = [R_{tot}] - [R^*]$ and setting $r = [R^*]$

$$k_R^+ a^2 ([R_{tot}] - r) = k_R^- i^2 r \implies r(k_R^+ a^2 + k_R^- i^2) = k_R^+ a^2 [R_{tot}] \quad (8)$$

Set $\beta = \frac{k_R^-}{k_R^+}$, we have

$$[R^*] = r = \frac{a^2}{a^2 + \beta i^2} [R_{tot}] \quad (9)$$

2. Positive feedback and oscillations

a. Assume that the concentrations c_1 and a reach rapid equilibrium. Set $\frac{da}{dt} = 0$, we have $a=c$. Hence

$$r = \frac{[R_{tot}]a^2}{a^2 + \beta i^2} = \frac{[R_{tot}]c^2}{c^2 + \beta i^2} \quad (10)$$

From $\frac{dc_1}{dt} = 0$, we have $c_1 = \frac{k_1 r}{\gamma_1 + k_0}$

$$\frac{dc}{dt} = k_0 c_1 - \gamma_0 c = \frac{k_0 k_1}{\gamma_1 + k_0} \frac{[R_{tot}]c^2}{c^2 + \beta i^2} - \gamma_0 c \quad (11)$$

Denote $V = \frac{k_0 k_1}{\gamma_1 + k_0} [R_{tot}]$, we have

$$c_0 = \frac{V}{\gamma_0} = \frac{k_0 k_1}{\gamma_0 (\gamma_1 + k_0)} [R_{tot}] \quad (12)$$

If we choose $\hat{t} = \gamma_0 t$, $\hat{c} = \frac{c}{c_0}$, $\hat{i} = \frac{i}{c_0}$, we have

$$\frac{d\hat{c}}{d\hat{t}} = \frac{\hat{c}^2}{\hat{c}^2 + \beta \hat{i}^2} - \hat{c} \quad (13)$$

$$\frac{d\hat{i}}{d\hat{t}} = \frac{k_I^-}{\gamma_0} (\hat{c} - \hat{i}) \quad (14)$$

i.e.

$$\frac{dc}{dt} = \frac{c^2}{c^2 + \beta i^2} - c, \frac{di}{dt} = k(c - i) \quad (15)$$

where $k = \frac{k_I^-}{\gamma_0}$.

b. Denote $f = \frac{c^2}{\beta i^2 + c^2} - c$, $g = k(c - i)$. By solving $f = 0$, $g = 0$, we can get the fixed point. $g = 0$ gives $i = c$, while $f = 0$ gives $\frac{1}{1+\beta} = c$ ($c \neq 0$). Hence the fixed point $c^* = i^* = \frac{1}{1+\beta}$.

Now we calculate the Jacobian matrix at the fixed point

$$\partial_c f = \frac{2c\beta i^2}{(\beta i^2 + c^2)^2} - 1 = \frac{\beta - 1}{\beta + 1}, \partial_i f = -\frac{2c^2\beta i}{(\beta i^2 + c^2)^2} = -\frac{2\beta}{\beta + 1}, \partial_c g = k, \partial_i g = -k \quad (16)$$

Hence the Jacobian matrix is

$$J = \begin{bmatrix} \frac{\beta-1}{\beta+1} & -\frac{2\beta}{\beta+1} \\ k & -k \end{bmatrix} \quad (17)$$

$\det J = k > 0$, $\text{tr} J = \frac{\beta-1}{\beta+1} - k$. The system is oscillatory if $\text{tr} J > 0$, which gives

$$k < \frac{\beta - 1}{\beta + 1} \quad \text{and} \quad \beta > 1 \quad (18)$$

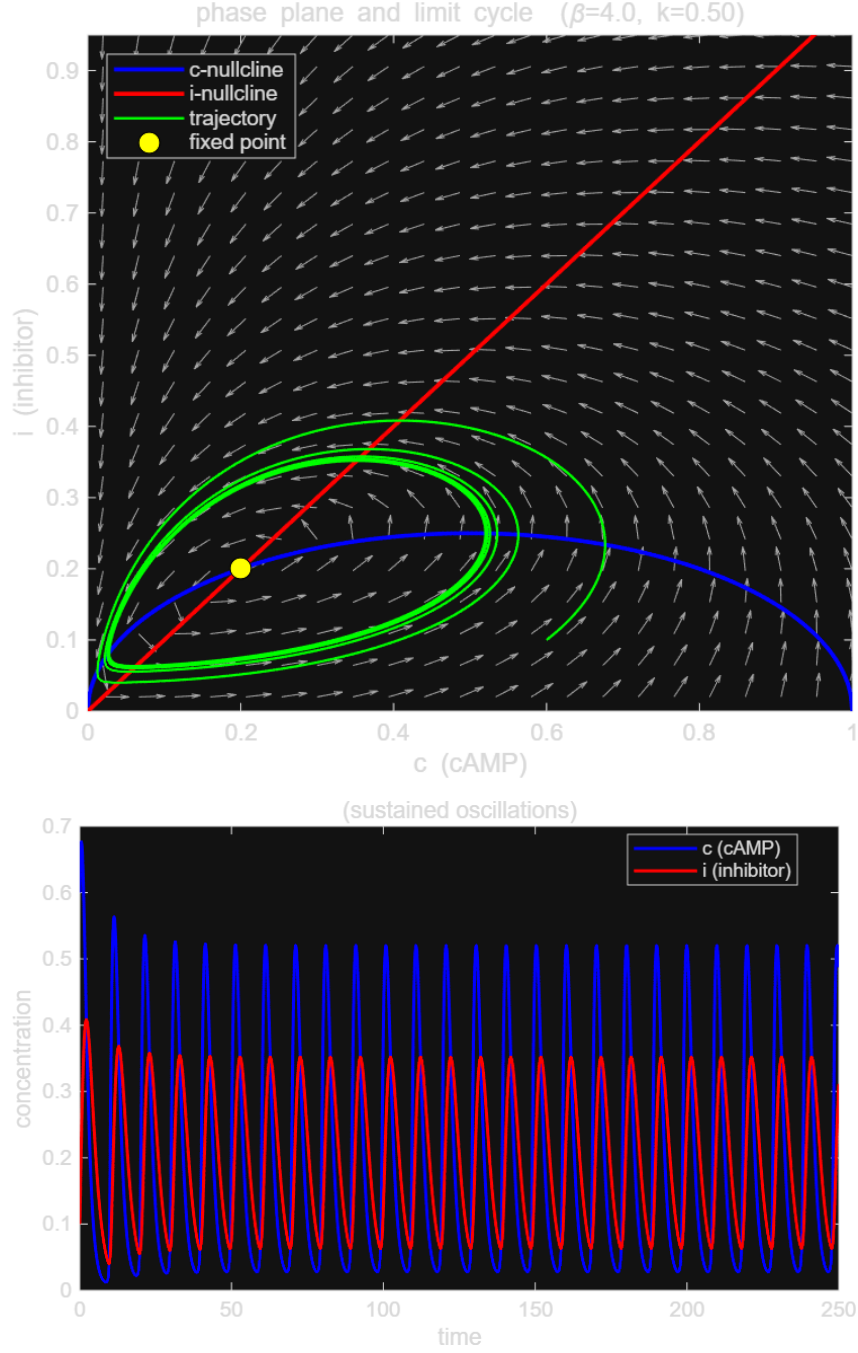
Remark: when $\text{tr} J > 0$, the positive fixed point becomes unstable. Since the concentrations remain positive and bounded, trajectories cannot diverge to infinity. Moreover, this is the only internal fixed point in the positive quadrant. Therefore, by the Poincaré - Bendixson theorem, trajectories are attracted to a closed orbit, corresponding to a stable limit cycle.

c. computation

We take $\beta = 4$, $k = 0.5 < k_c = 0.6$

c-nullcline: $i = \sqrt{\frac{c(1-c)}{\beta}}$, i-nullcline: $i = c$. The intersection of the two nullclines is the fixed point $(c^*, i^*) = (0.2, 0.2)$, which lies on the ascending branch of the c-nullcline. From part (b), the trace at this point is positive, so the fixed point is unstable. As a result, trajectories spiral outward and are eventually attracted to a limit cycle.

The time series shows that c and i exhibit stable, phase-lagged sustained oscillations, with i lagging behind c. This reflects a “delayed negative feedback” mechanism and corresponds to a limit cycle.



d. Suppose the system only required a single molecule of AC or IC.

$$\frac{dc}{dt} = \frac{c}{c + \beta i} - c, \quad \frac{di}{dt} = k(c - i) \quad (19)$$

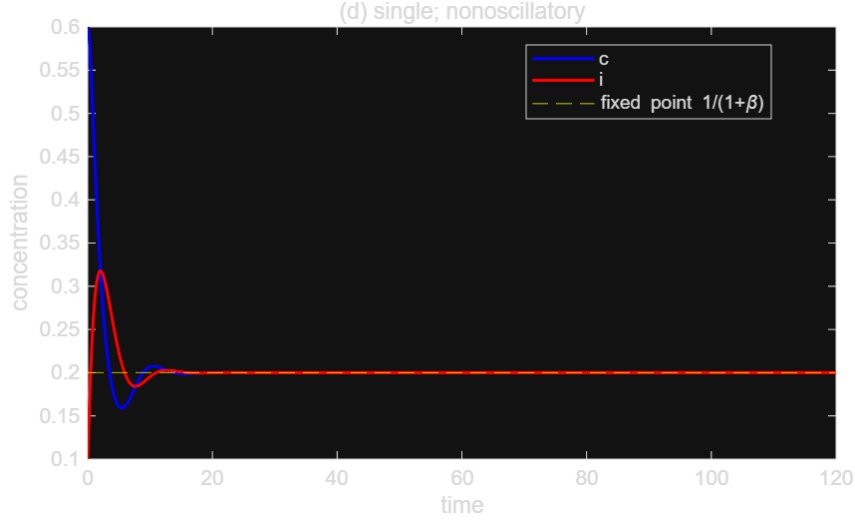
The fixed point is still $c^* = i^* = \frac{1}{1+\beta}$. The Jacobian matrix at the fixed point is

$$J = \begin{bmatrix} \frac{-1}{\beta+1} & -\frac{\beta}{\beta+1} \\ k & -k \end{bmatrix} \quad (20)$$

$$\det J = k > 0, \text{tr} J = -\frac{1}{\beta+1} - k < 0.$$

The trace is always negative, so the internal fixed point is always stable and the system cannot oscillate. For any initial condition, the trajectory decays back to the steady state. Below is an

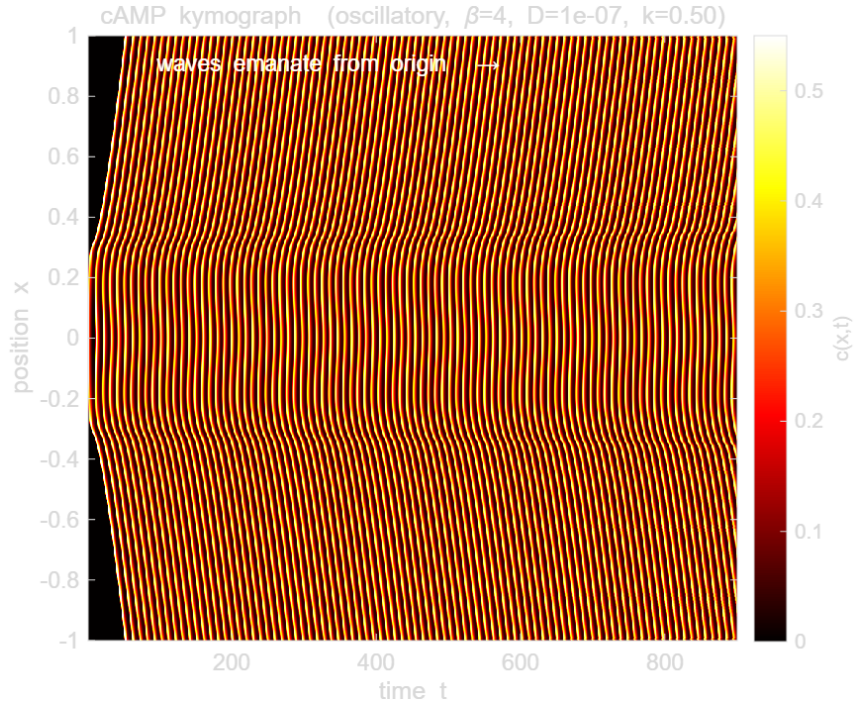
example. This indicates that cooperativity or ultrasensitivity, which requires paired molecules, is a necessary condition for generating oscillations. The nonlinear steepness provided by a^2 and i^2 makes the positive feedback strong enough to destabilize the steady state and form a limit cycle. Under single-molecule, linear catalysis, the feedback is too gradual, and the oscillations disappear.



3. Diffusion and cAMP waves

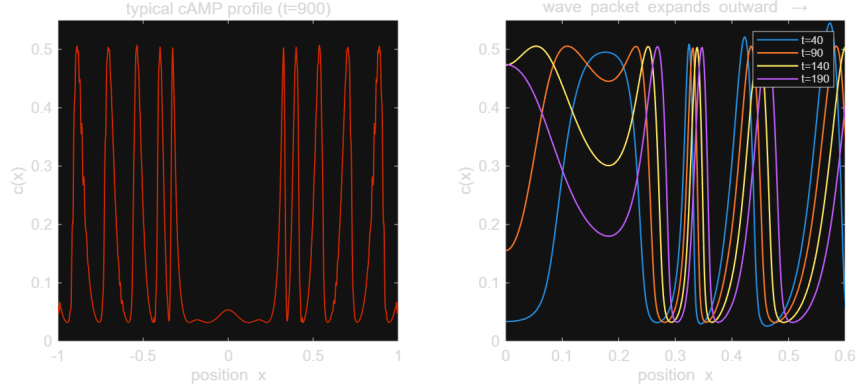
a. $\beta = 4, k = 0.5, D = 10^{-7}, \sigma = 0.1, i_0 = 0.1$

The simulation produces an oscillatory cAMP field. In the kymograph, the horizontal axis is time t , the vertical axis is position x , and the color represents the cAMP concentration $c(x, t)$.



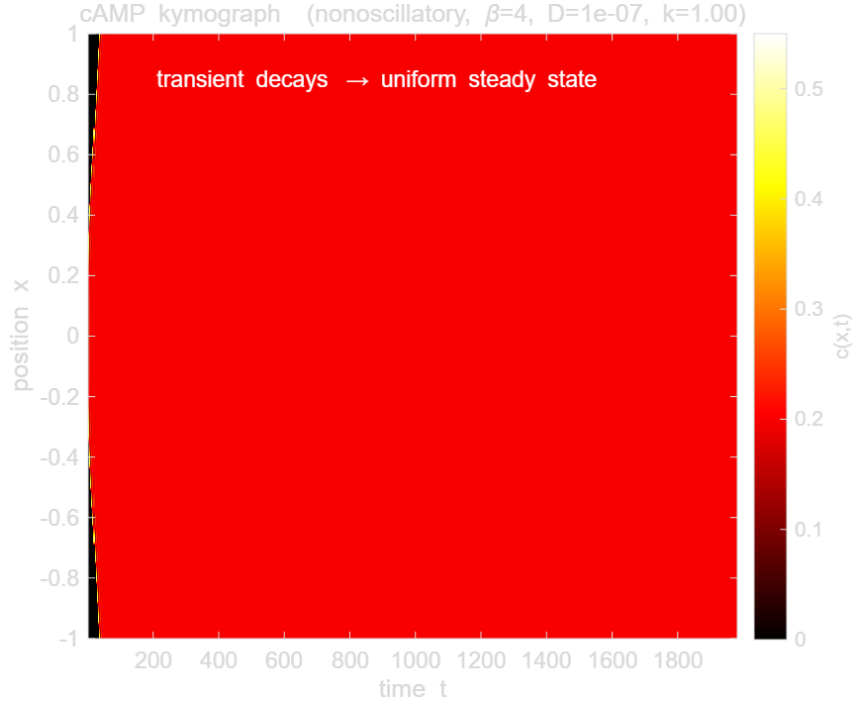
The kymograph shows repeated bright diagonal stripes starting near $x = 0$, which correspond to cAMP wave fronts. Since the bands move away from the origin as time increases, the result shows that cAMP waves are emitted from the origin and propagate outward.

A typical spatial cAMP profile at $t=900$ contains a train of sharp cAMP peaks, which represent individual wave fronts. On the right side of the origin, the peaks move toward increasing x , while on the left side they move toward decreasing x .



Therefore the direction of wave propagation is away from the origin, which is consistent with the kymograph. The oscillating system does not settle to a static gradient. Instead, it produces repeated pulses of cAMP that travel away from the source.

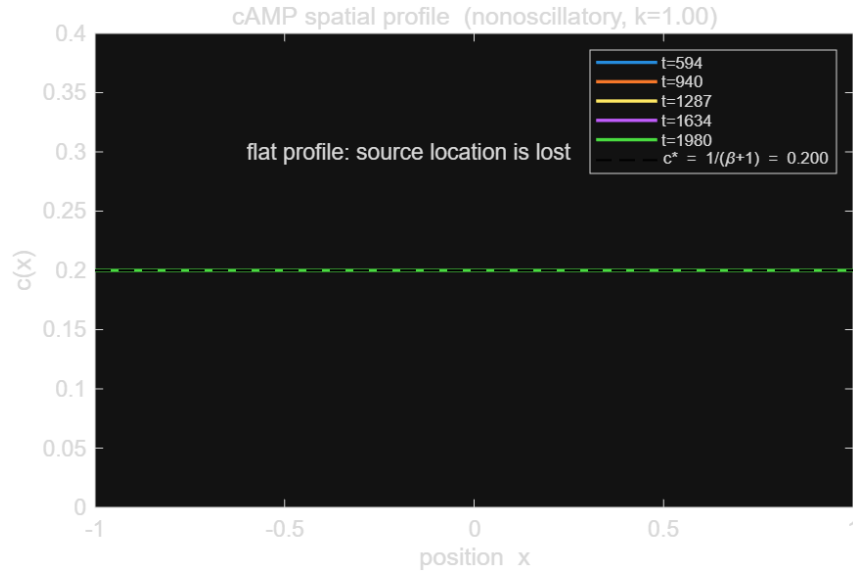
b. To obtain a non-oscillating system, I increase k to $k = 1.0$. Any initial transient behavior quickly dies out, and the whole spatial domain approaches a uniform steady state.



The typical cAMP profile after the transients have disappeared is flat: $c(x, t) \approx c^*$. The steady-state value is

$$c^* = \frac{1}{1 + \beta} = \frac{1}{5} = 0.2 \quad (21)$$

Thus after sufficient time, the profile becomes approximately $c(x, t) \approx 0.2$



Since the profile is spatially uniform, there is no concentration gradient or remaining information about where the original cAMP source was located. Cells cannot determine the initial source location from this type of profile. The source information has effectively been lost.

c. For a static cAMP gradient, the signal would be produced mainly by diffusion from the source. Because cAMP is also degraded by phosphodiesterase, the concentration would decrease rapidly with distance. Far from the source, the concentration difference across a cell body would become so small that the signal-to-noise ratio would be too low for reliable chemotaxis.

In contrast, cAMP waves are regenerative. When cells are stimulated by cAMP, they actively amplify the signal and secrete more cAMP themselves. Thus the wave is relayed from cell to cell, so the signal can propagate over a much longer distance than a simple passive gradient.

Waves also avoid receptor saturation. If a static gradient were made strong enough to be detected far away, the cAMP concentration near the source would likely be so high that receptors would saturate, reducing directional sensitivity. A pulsed wave allows cAMP to rise transiently and then fall again, keeping cells within a useful dynamic range.

Finally, repeated waves provide reliable directional cues and synchronize the population. Cells respond mainly to the rising front of each wave and then adapt after the wave passes. Since waves propagate outward from the aggregation center, the spatial gradient at the wave front points back toward the source, allowing cells to move step by step toward the center. Therefore, cAMP waves are more robust and effective than a static gradient for coordinating large-scale aggregation in *Dictyostelium*

Appendix

MATLAB code for 2(c)(d)

```
function ex2_c

clear; close all; clc;

beta = 4;
```

```

k      = 0.5;
kc     = (beta-1)/(beta+1);
fprintf('Hopf threshold  k_c = (beta-1)/(beta+1) = %.4f\n', kc);
if beta>1 && k<kc
    fprintf('beta=%.2f, k=%.2f < k_c -> oscillatory\n',beta,k);
else
    fprintf('beta=%.2f, k=%.2f -> nonoscillatory\n',beta,k);
end

f = @(c,i) c.^2./(beta*i.^2 + c.^2) - c;
g = @(c,i) k*(c - i);
cstar = 1/(1+beta);  istar = cstar;
fprintf('fixed point: c* = i* = %.4f\n', cstar);

ode = @(t,y) [f(y(1),y(2)); g(y(1),y(2))];
opts = odeset('RelTol',1e-9,'AbsTol',1e-11);
[t,Y] = ode45(ode,[0 250],[0.6;0.1],opts);

cc = linspace(1e-4,1,500);
i_cnull = sqrt(max(cc.*(1-cc),0)/beta);
i_inull = cc;

figure('Color','w','Position',[80 80 560 470]); hold on; box on;
[Cg,Ig] = meshgrid(linspace(0.02,1,22),linspace(0.02,0.95,22));
U=f(Cg,Ig); V=g(Cg,Ig); Nn=hypot(U,V)+eps;
quiver(Cg,Ig,U./Nn,V./Nn,0.55,'Color',[.65 .65 .65]);
p1=plot(cc,i_cnull,'b-','LineWidth',1.8);
p2=plot(cc,i_inull,'r-','LineWidth',1.8);
p3=plot(Y(:,1),Y(:,2),'g-','LineWidth',1.1);
p4=plot(cstar,istar,'ko','MarkerFaceColor','y','MarkerSize',9);
xlabel('c (cAMP)'); ylabel('i (inhibitor)');
title(sprintf('phase plane and limit cycle (\\beta=%.1f, k=%.2f)',beta,k));
legend([p1 p2 p3 p4],{'c-nullcline','i-nullcline','trajectory','fixed
↪ point'},'Location','northwest');
axis([0 1 0 0.95]);
savefig_safe('fig_p2_phase.png');

```



```

figure('Color','w','Position',[660 80 620 360]); hold on; box on;
plot(t,Y(:,1),'b-','LineWidth',1.5);
plot(t,Y(:,2),'r-','LineWidth',1.5);
xlabel('time'); ylabel('concentration');
legend('c (cAMP)','i (inhibitor)','Location','best');
title('(sustained oscillations)');
savefig_safe('fig_p2_time.png');

f1 = @(c,i) c./(beta*i + c) - c;
ode1 = @(t,y) [f1(y(1),y(2)); g(y(1),y(2))];
[t1,Y1] = ode45(ode1,[0 120],[0.6;0.1],opts);

figure('Color','w','Position',[300 470 620 340]); hold on; box on;
plot(t1,Y1(:,1),'b-','LineWidth',1.5);
plot(t1,Y1(:,2),'r-','LineWidth',1.5);
yline(1/(1+beta),'y--');
xlabel('time'); ylabel('concentration');
legend('c','i','fixed point 1/(1+\beta)','Location','best');
title('(d) single; nonoscillatory');
savefig_safe('fig_p2_single.png');

fprintf('\n(d) single molecule Jacobian trace = -1/(1+beta) - k = %.4f < 0 ->
↪ nonoscillatory\n',...
        -1/(1+beta)-k);
end

function savefig_safe(fname)
try
    exportgraphics(gcf,fname,'Resolution',150);
catch
    saveas(gcf,fname);
end
end

```

MATLAB code for 3

```
function ex3
```

```
clear; close all; clc;
```

```
beta = 4;
```

```

D      = 1e-7;
sigma = 0.1;
i0     = 0.1;

cstar = 1/(beta+1);
kc     = (beta-1)/(beta+1);
fprintf('Steady state c* = %.4f , Hopf threshold k_c = %.4f\n', cstar, kc);

N = 1601; x = linspace(-1,1,N)'; dx = x(2)-x(1);

cases = struct( 'name', {'oscillatory'      , 'nonoscillatory' }, ...
                'k'   , {0.5                , 1.0                }, ...
                'T'   , {900                 , 2000                }, ...
                'dt'  , {0.05                , 0.05                }, ...
                'nSnap',{900                 , 600                 });

for ci = 1:numel(cases)
    k = cases(ci).k; nm = cases(ci).name;
    T = cases(ci).T; dt = cases(ci).dt; nSnap = cases(ci).nSnap;
    nT = round(T/dt); snapEvery = max(1,floor(nT/nSnap));
    fprintf('\nRunning %-14s : k = %.2f (Hopf threshold %.3f)\n', nm, k, kc);

    if dt > dx^2/(2*D)
        warning('dt=%.3g exceeds diffusion stability limit %.3g', dt, dx^2/(2*D));
    end

    c = exp(-x.^2/(2*sigma^2));
    i = i0*ones(N,1);

    Csnap = zeros(N,nSnap); tSnap = zeros(1,nSnap); s = 0;

    lap = zeros(N,1);
    for n = 1:nT
        lap(2:N-1) = (c(3:N) - 2*c(2:N-1) + c(1:N-2))/dx^2;
        lap(1)     = 2*(c(2) - c(1))/dx^2;
        lap(N)     = 2*(c(N-1) - c(N))/dx^2;

        fc = c.^2./(beta*i.^2 + c.^2) - c;
        c = c + dt*(fc + D*lap); c = max(c,0);
    end
end

```

```

    i = i + dt*(k*(c - i));

    if mod(n,snapEvery)==0 && s<nSnap
        s = s+1; Csnap(:,s) = c; tSnap(s) = n*dt;
    end
end
Csnap = Csnap(:,1:s); tSnap = tSnap(1:s);

figure('Color','w','Position',[80 80 600 480]);
imagesc(tSnap, x, Csnap); set(gca,'YDir','normal');
colormap(hot); caxis([0 0.55]);
xlabel('time t'); ylabel('position x');
cb = colorbar; cb.Label.String = 'c(x,t)';
set(gca,'FontSize',11,'Layer','top');
title(sprintf('cAMP kymograph (%s, \beta=%g, D=%g, k=%.2f)', nm, beta, D,
    ↪ k));
if strcmp(nm,'oscillatory')
    text(0.10,0.95,'waves emanate from origin \rightarrow', ...
        'Units','normalized','Color','w','FontSize',11);
else
    text(0.10,0.92,'transient decays \rightarrow uniform steady state', ...
        'Units','normalized','Color','w','FontSize',11);
end
exportfig(sprintf('fig_p3_%s_kymo.png', nm));

if strcmp(nm,'oscillatory')
    figure('Color','w','Position',[700 80 980 380]);

    subplot(1,2,1); box on;
    plot(x, Csnap(:,end), 'LineWidth',0.9, 'Color',[0.85 0.16 0.0]);
    ylim([0 0.55]); xlabel('position x'); ylabel('c(x)');
    title(sprintf('typical cAMP profile (t=%.0f)', tSnap(end)));
    set(gca,'FontSize',11);

    subplot(1,2,2); hold on; box on;
    te = [40 90 140 190];
    je = arrayfun(@(tt)find(abs(tSnap-tt)==min(abs(tSnap-tt)),1), te);
    plot(x, Csnap(:,je), 'LineWidth',1.1);
    xlim([0 0.6]); ylim([0 0.55]);
    xlabel('position x'); ylabel('c(x)');
    title('wave packet expands outward \rightarrow');

```

```

legend(arrayfun(@(tt)sprintf('t=%.0f',tt),tSnap(je),'Uni',0), ...
        'Location','northeast','FontSize',8);
set(gca,'FontSize',11);

exportfig(sprintf('fig_p3_%s_profile.png', nm));

else
    figure('Color','w','Position',[700 80 640 380]); hold on; box on;
    idx = round(linspace(round(0.30*s), s, 5));
    plot(x, Csnap(:,idx), 'LineWidth',1.6);
    plot([-1 1],[cstar cstar],'k--','LineWidth',1);
    ylim([0 0.4]); xlabel('position x'); ylabel('c(x)');
    title(sprintf('cAMP spatial profile (%s, k=%.2f)', nm, k));
    lg = arrayfun(@(tt)sprintf('t=%.0f',tt),tSnap(idx),'Uni',0);
    lg{end+1} = sprintf('c* = 1/(\beta+1) = %.3f', cstar);
    legend(lg,'Location','northeast','FontSize',8);
    text(-0.6,0.30,'flat profile: source location is lost','FontSize',11);
    set(gca,'FontSize',11);
    exportfig(sprintf('fig_p3_%s_profile.png', nm));
end
end

fprintf('\nDone. Figures saved as fig_p3_*.png\n');
end

function exportfig(fname)
try
    exportgraphics(gcf, fname, 'Resolution', 150);
catch
    print(gcf, fname, '-dpng', '-r150');
end
end

```